Nicole Hee-Yeon Kim

nicole2kim@gmail.com | Personal Page
Nationality: United States of America and Republic of Korea (Dual Citizenship)

Research Interest

Efficient and Scalable AI Systems; Optimization for Large-Scale Deep Learning Models; Training and Inference Efficiency in LLMs and VLMs; Robust and Reliable AI

Education

- **Korea Advanced Institute of Science and Technology** — *Feb 2024 - Mar 2026*
  Master's Program, Department of Industrial & Systems Engineering, Overall GPA: 3.95/4.3 — Daejeon, South Korea
- **Yonsei University** — *Mar 2019 - Feb 2024*
  Bachelor's Program, Department of Industrial Engineering, Overall GPA: 3.77/4.3 — Seoul, South Korea

Publications

1. Lee, Y.*, Shin, J.*, **Kim, N.***, Bang, J., Lee, J., Hwang, K., Porikli, F., Song, H. (2026). Rethinking RAG in Long Videos: What to Retrieve and How to Use It?. *Submitted to NeurIPS 2026 (Average score: 4/6)*
(* Equal contribution.)

2. **Kim, N.** and Song, H. (2025). Robust Dataset Condensation using Supervised Contrastive Learning. In *Proceedings of International Conference on Computer Vision (ICCV 2025)*, Main Conference Paper.

3. **Kim, N.**, Choi, J., Lee, Y., Song, H. (2025). Robust Dataset Condensation via Semi-Supervised Learning. In *Proceedings of the Korea Computer Congress (KCC 2025)*, Selected for Oral Presentation.

4. **Kim, N.**, Lee, Y., Song, H. (2024). Robust Dataset Condensation via Supervised Contrastive Learning. In *Proceedings of the Korea Software Congress (KSC 2024)*, Selected for Oral Presentation.

5. Ban, M., Choi J., Min H., **Kim, N.**, Kim M., Lee J., Song H. Completing Missing Annotation: Multi-Agent Debate for Accurate and Scalable Relevant Assessment for IR Benchmarks. In *Proceedings of International Conference on Learning Representations (ICLR 2026)*.

6. Lee, Y., Deng, J., **Kim, N.**, Min, H., Yun, T., Ban, M., Song, H. (2025). Towards a Holistic and Automated Evaluation Framework for Multi-Level Comprehension of LLMs in Book-Length Contexts. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP 2025)*, Main Conference Paper.

7. Min, H., Lee, Y., Ban, M., Deng, J., **Kim, N.**, Yun, T., Su, H., Cai, J., Song, H. (2025). Towards Multi-dimensional Evaluation of LLM Summarization across Domains and Languages. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (ACL 2025)*, Main Conference Paper.

8. Kim, T., Lee, S., Kang, J., Choi, Y., Yun, W., **Kim, N.**, Chen, Z., Xie, L., Song, K. (2025). IMC: A Benchmark for Invariant Learning under Multiple Causes. In *Proceedings of the CVPR 2025 Workshop on Domain Generalization: Evolution, Breakthroughs, and Future Horizons*, Best Paper Award.

9. Oh, J., Choi, J., **Kim, N.**, Yun, T., Kwon, R., Song, H. (2025). Learning to Verify Summary Facts with Fine-Grained LLM Feedback. In *Proceedings of the 2025 International Conference on Computational Linguistics (COLING 2025)*, Selected for Oral Presentation.

10. Oh, J., Choi, J., **Kim, N.**, Song, H. (2025). Improving Language Model Quality through LLM-based Fine-Grained Hallucinated Summary Generation. *Journal of Computing Practice*, vol. 31(2), pp. 91-97.

11. Oh, J., Choi, J., **Kim, N.**, Song, H. (2024). Improving the Text Summary Quality Through Understanding the Hallucination Level of Summarization Using Large Language Models. In *Proceedings of the Korea Computer Congress (KCC 2024)*, Selected for Oral Presentation.

Research Experience

- **DISLab, Department of Industrial & Systems Engineering, KAIST** *Feb 2024 - Present*
  Master's Student Daejeon, South Korea
  - Led the research project “Robust Dataset Condensation using Supervised Contrastive Learning“ as the sole first author (accepted at ICCV 2025).
  - Led the research project “Qualcomm Industry–Academia Collaboration Project“ titled “Rethinking RAG in Long Videos: What to Retrieve and How to Use It? “.
  - Participated in research project “Development of VLM-based PDF Document Understanding and Automated Road Alignment Coordinate Extraction Technology“, supported by ETRI.
- **MLAI Lab, Department of Applied Statistics, Yonsei University** *Sep 2023 - Feb 2024*
  Undergraduate Research Intern Seoul, South Korea
  - Designed and constructed the PlayingCard-10 dataset, a core component of "IMC: A Benchmark for Invariant Learning under Multiple Causes" (CVPR 2025 Workshop Best Paper Award).

Patent

- Robust Dataset Condensation using Supervised Contrastive Learning (2025, Pending).
- Cost-efficient and Accurate Debate-based Relevance Assessment System with Multi-agents(2025, Pending).

Scholorships

- **KAIST Support Scholarship** *Feb 2024 - Feb 2026*
  Government-funded full tuition scholarship for M.S. program KAIST
- **Brain Korea 21 (BK21) Scholarship** *Feb 2024 - Feb 2026*
  Government-funded research scholarship for graduate students National Research Foundation of Korea
- **Yonsei Welfare Scholarship** *Mar 2019 - Jun 2023*
  Full tuition scholarship for B.S. program Yonsei University
- **University Innovation Support Scholarship** *Sep 2020 - Feb 2021*
  Scholarship awarded for the development and advancement of an innovative start-up idea Yonsei University

Honors and Awards

- **Best Paper Award** *Jun 2025*
  CVPR 2025 Workshop on Domain Generalization: Evolution, Breakthroughs, and Future Horizons
- **Outstanding Presentation Paper Award** *Jul 2024*
  Korea Computer Congress 2024
- **1st Place, Promotional Video Contest** *Feb 2022*
  Department of Industrial Engineering, Yonsei University
- **Yonsei Social Entrepreneurship Award** *Feb 2021*
  Yonsei University

Language Proficiency

- **Languages:** Korean (Native), English (Fluent, TOEFL iBT: 111 [R30 L27 S28 W26])

Teaching Experience

- **Yonsei–NAVER Cloud Data Science Program** *Dec 2020 - Dec 2023*
  Assisted in designing, editing, and organizing instructional materials and coding sessions Yonsei University

Leadership Experience

- **Academic & Campus Life Mentor for Freshmen** *Oct 2019 - Aug 2020*
  Institute for Higher Education Innovation, Yonsei University
- **Freshman Class President** *Mar 2019 - Aug 2019*
  Department of Industrial Engineering, Yonsei University